



Mixed initiative collaboration: the missing agency in LLMs

- define a probe: mechanistic/verbal
 - mechanistic: logistic regression on hidden state representation
 - verbal: presupposition (hypo: difference in behavior between "solve the problem" vs. "solve the problem if possible")
- train on self-reflection
- define a dataset that we want to use: look at QuestBench (but we also need the complete version, so we should check whether they have a 1-1 matching)
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- RLHF data from Anthropic (2022): <https://huggingface.co/papers/2204.05862>
 - OLMO Allen AI data: <https://huggingface.co/allenai>
 - <https://huggingface.co/collections/allenai/olmo-3-post-training>
 - DCLM: <https://huggingface.co/apple/DCLM-7B>
 - alpaca: <https://huggingface.co/datasets/tatsu-lab/alpaca>
 - Post training collection: <https://github.com/mlabonne/llm-datasets>

Personalization effectively optimizes for the speakers objective function but that may end up harming the objective function of the collective (society). So in a way you want to find the Pareto optimality between speakers objective function and the objective function of the collective.

Our main problem statement:

1. Current preference pairs are listener centric.
2. These listener centric preference pairs remove the communicative process.
3. We propose an alternate way to gather preference data i.e. speaker centric.
4. We hypothesize that this speaker centric preference pairs will embody the communicative process.
5. We hypothesize that phenomenons like sycophancy, lack of appropriate clarifying questions, etc. all are a result of the communicative process being removed and a model trained on the speaker centric preference pairs is able to solve for these phenomenons.
 - a. In a sense speaker-centric preference pairs remove the model from the *listener-centric bubble*. It is in a way giving all of humanity a voice while taking to a particular listener. It does not overfit to a single agent (listener) but all the speakers (humanity) instead.

Limitations:

1. Persuasion

Bottlenecks:

1. ~~Neurips~~
2. Hypothesis A: speaker centric preference pairs will embody the communicative process
 - a. A speaker centric preference pair is where a speaker is presented with the listeners request without context about the listener. The speaker has to then choose preference from a pair or generate an utterance that best matches the action they would take next.
 - i. If we can engineer a system that can present preference pairs that align closely with the speakers actions then we can allow the speaker to choose a response.
 1. What is the action space of the speaker?
 - ii. To generate an utterance the speaker needs to know the domain or be an expert in it.
 1. Or they can use an LLM to bridge that gap but how do we do that?
3. Hypothesis A':
 - a. Rationale: So we have the blockers of not knowing the preference pairs and we don't know how we can create an LLM to help bridge the knowledge gap of the

- speaker.
- b. So what can we do? Let's say we start with a problem in LLM that best represents the lack of communicative process.
 - c. **Hypothesis:** LLMs have a verbalization gap, i.e. they do not verbalize what they already know (know more than they can say)
 - d. What are types of things that LLMs may know but don't say?
 - i. insufficient info (corresponding communicative behavior: asking clarifying questions)
 - ii. uncertainty about their own knowledge (corresponding communicative behavior: hedging)
 - iii. ambiguity in the instruction (corresponding communicative behavior: asking clarifying questions)
 - iv.
 - e. Test this on domains: [Math/Reasoning](#), LLMs get lost in multi turn
4. **Hypothesis B:** Training on synthetic preference pairs of model expressing itself allows model to close the verbalization gap?
- a. Step 1: We gather preference pairs from the above domains where the model expresses itself vs it does not (baseline).
 - b. Step 2: We do RL on a model: one on baseline and another on the expressive preference data.
 - c. Step 3: Did our model generate utterances that are better aligned to its internal representation.
5. **Hypothesis C:** Models that are speaker-centric have downstream effects communicative process?
- a. We measure communicative process through the following phenomenon:
 - i. Sycophancy
 1. <https://arxiv.org/pdf/2502.08177v4>
 2. <https://arxiv.org/pdf/2310.13548>
 - a. This paper has a dataset for evaluating sycophancy
 - b. They also use (bayesian) logistic regression to understand what features do human raters prefer
 - c. Then, they also evaluate from a third party to see whether humans prefer truthful helpful answers or sycophantic answers without giving them access to ground truth and internet
 - ii. Asking Clarifying Questions
 - iii. Extreme Political Views
 - iv. Expression of Uncertainty
 - v. Verbosity
 - vi. truthful vs helpful
 - vii. Grounding Gaps
 - viii. Multi Turn Performance?????
 - ix. social reasoning:
 1. <https://aclanthology.org/2024.eacl-long.138/>
 2. <https://www.nature.com/articles/s43588-024-00741-1>

Hypos

A: LLMs do not verbalize what they already know

B: LLMs do not reject presuppositions

C: A happens because of B

One way of showing this is: if LLMs are allowed to reject presuppositions, then they do verbalize what they know. [counterfactual intervention]

V2:

1. **Hypothesis A:** LLMs do not verbalize what they already know (know more than they can say)
2. Validate the ^ hypothesis in one or more settings. Need to show two things:
 - a. we know that the request is unfulfillable, and (the probe tells us that) the model also knows
 - b. the model does not verbalize it in its response
 - c. Action plan: for each unfulfillable request
 - i. does the model know? (via probing)
 - ii. if i is yes, does the model verbalize it?
3. [??] Create preference pairs based on the settings in which hypothesis is validated.
- 4.
5. Test the two models
6. See if the training generalizes to the phenomenon
 - a. Sycophancy

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7. Human eval?

V2:

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3. **Hypothesis B:** LLMs do not verbalize what they already know because all else equal, a response that shows “lack of agency” is preferred.
4. Hypothesis C: LLMs lack the agency to verbalize what they know because of the user-centric way of gathering preference data right now.
5. Validate the above ^ hypothesis by training two models one trained on user-centric preference pairs vs H2 preference pairs.
6. [??] Create preference pairs based on the settings in which hypothesis is validated.
- 7.
8. Test the two models
9. See if the training generalizes to the phenomenon
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10. Human eval?

Alternate progress:

1. AI lack agency → is it because the model is overfitting?

2. Evaluate lack of agency
3. Show that it is because of the framing gap
4. Train the model

Phase 1:

1. What phenomenon are we studying?
 - a. What is the model doing (behavior)
 - b. What is the postulated origin of the behavior?
 - c. What is the effect observed in human annotations (if available)
2. What evaluation metrics are available to measure these phenomenon?
3. Pick a domain - Information Seeking
4. Look for post training datasets — when we find a dataset, does it have all the info we need to answer the questions above?
5. Evaluate the post training dataset using the evaluation metrics for the different phenomenon

Phase 2:

1. Put down the hypothesis: Unify the phenomenon as a result of the framing gap
 - a. What does the framing gap do to annotations?
 - b. What is the different between self talk vs talking to other?

State that the self talk or the egocentric view of talking removes agency from the model

Phase 3:

1. Create a new dataset that is from the new framing of annotation
2. Evaluate the dataset using the metrics to see the differences for the selected phenomenon
3. Train a model on the new dataset
4. Evaluate the model on the selected phenomenon

ScratchPad

📅 Friday, March 13th

pitch:

1. models don't talk like humans; for example,
 - a. they are bad at conveying uncertainty,
 - b. they go on forever before asking simple clarifying questions (which would save both inference costs and user time),
 - c. they are too easily persuaded
2. NLP folks have tried to tackle these separately
 - a. [wrt 1a] <https://arxiv.org/abs/2405.21028> (elias)
 - b. [wrt 1b]
 - i. <https://arxiv.org/pdf/2410.13788> (eunsol): need for asking clarifying question arises from ambiguity in the user request semantics), e.g. `who's the highest-paying football player`
 - ii. STaR-GATE (Noah Goodman); need for asking clarifying question arises from underspecification of user persona, e.g. `What is a good pasta recipe?` (later finds out user is vegetarian)
 - c. [wrt 1c] <https://arxiv.org/abs/2410.14596> (elias)
3. we claim that there exist a unifying framing gap (motivated by pragmatics)

evaluation idea: evaluate on RIFTS, *and* these tasks separately

Other related lit: RLHS

Pipeline for the new paper:

- Gather papers and their prompting strategies
- Look at the undesirable model behavior that they are talking about

- Look at what the prompting strategies are trying to achieve and how are they steering the model
- Boil it down to **the framing gap in LLM training**
- Gather their datasets (Create a held out and eval test)
- Generate the H1 and H2 responses to the train set
- Train 2 models (one on H1 and one on H2)
- Eval the model on held out set
 - H1 is a baseline for all settings
 - for specific phenomena, we will have additional baselines (from past work)
 - goal: $H2 > H1$ for all settings; H2 rivals phenomenon-specific finetuning in a particular given setting

goal: define collaboration, operationalize it in terms of supervision collection

Nikhil Sharma 11:33 PM

November 19th, 2025 ▾

@channel Look at the oasst1 dataset. Just look at the question, try to come up with a response that you think you would say if someone said that to you and compare it with the answer in the dataset.

I am looking at every paper since. I cannot believe people didn't see this. I think i might pass out.

1

I cannot believe we have wasted so much compute on problems that could have been fixed with such simple reframing

Look at Llama3 paper. Go over the section of hallucination and look at what they do. This is a made up problem with a madeup solution. Now if you think about it the H2 framing already addresses this or repositions this problem as well.

Basically they keep interrogating the model using judges and based on variance they figure out if the model know or doesn't know

And then using generated examples from this synthetically generated conversation they can be trained to make inferences based on their known knowledge. Under the hood it would learn if a particular neuron of "uncertainty" is high then I don't answer the question. (I here is the model)

Nikhil Sharma 11:33 PM

What happens in reinforcement learning is the model generates 1000s of answers on perturbation of questions (usually these are math and computation heavy questions). We prune the answers that were wrong and then train the model on the selected response.

What happens here is actually correct. Since there is no human involved the training actually teaches the model to focus on what works for itself.

Can we convert H1 to H2?

Why we collaborate:

What should be the goal (i.e. optimization objective) of a collaborative AI agent:

- status quo: expected utility of H2's decision in place of H1
- want: expected utility of H1's decision with *input* from H2

This encompasses aligning user's self-perceived utility function with the (unknowable) ground-truth utility function.

- e.g., when the user doesn't even know what their goal is or what's good for their goal, the agent and the user need to collaborate to sharpen the goal.

aside: there are two sources of error --

- 1) error in utility function estimate
- 2) choosing optimal action based on utility function

Delegation:

- Starts with a pre defined goal

What are the emergent behaviors of collaboration:

- power to steer the conversation (act to steer the conversation)
 - repair [rifts paper]

- joint attention
- In second person not 3rd person i.e. as a interaction partner not as an object of observation
- Both agents should have access to their own goals and each others goals through conversation
- Emergent goal - not a goal to begin with (that is delegation)

Codebook

Dataset: (show humans are doing something that AI is not doing)

- WildChat
- Do we have human human collaboration dataset?

Roadmap:

- whether or not a task requires collaboration is itself a classifier based on the first turn of a conversation
 - validate: success rate in human-AI logs correlates with delegation
- for the collaborative tasks, come up with a taxonomy of emergent behaviors
 - use human-human logs as baseline to establish validity of taxonomy
 - use human-AI logs to show that AI is failing to do these
- [[@Fei Fang](#)] Operationalize the objective function - HOW
- case study (deeper dive on a particular task)
- TBD

Conclusion as of 📅 Wednesday, February 4th:

Fei's framing: decision-oriented

- H1: be a good decision maker in place of the user (conflating user with assistant)
- H2: make decisions that help the user be a good decision maker (recognizing user and assistant as separate entities)

Nikhil's interpretation/alternative framing:

- H1: make decisions → self-talk
- H2: ...help user be good decision maker → communicating with user

📅 Friday, February 27th

there are two axes:

1. single turn vs multi turn
2. delegation vs collaboration

out of the 2x2 cartesian product, single-turn cannot co-exist with collaboration. so we've got:

single-turn delegation ➡ multi-turn delegation



multi-turn collaboration

in terms of annotation collection: multi-turn delegation and multi-turn collaboration *could* arise from the same annotation prompt (e.g. this other person has come to you for help). whether or not we are in the delegation or collaboration bucket may be inferred by the annotator from the user prompt (e.g. let's build a website vs build a website for me).

Next wed: start designing annotation task. we can fill in the taxonomy below in parallel.

Delegator same in single turn and multi turn

Delegatee

Need to be orthogonal

completion - delegation - collaboration

Map conversation to the above definition:

- 1+1 = completion

In HAI collaboration most are collaboration

What are the desired behavior - power dynamics (negotiate power in collaboration)

Magically find a dataset and fine tune the models to collaborate
comparision to dataset
distribution of task and interaction to match
10000 - 30000

DATASET COLLECTION:

Multi-turn delegation:

- No external tools required:

Questions from delegatee | Answers from delegator | Draft from delegatee _if we have trace on how the user came up with the draft that would be awesome (think aloud)) | Review from delegator

- Co-writing (email, wikipedia, etc.):
- Turning notes into presentation

- Tools required:

Tool calls*

- PM/tech lead - engineer
- Information seeking
- E-commerce

Collaboration:

- No external tools:

Questions | Answers | Edits |Comments

- Writing an email:
- Co-writing:
- Turning notes into presentation:

- External tools:

Tool calls

- Pair programming
- Information seeking
- E-commerce
- Open source coding (Github)

	Completion	Multi-Turn Delegation	Multi-turn Collaboration
Definition	well defined: no affordance - maximum likelihood estimate Theory of mind	well defined: clarrification affordance - preferece elicitation	ill defined both - process: collaborate
Requirements	- Hierarchy → delegatee's agency is at the discretion of the delegator - Hierarchy, continued: Accountability with the delegator - sequential attention - Existing common ground being aligned → Existing Goal + existing information needed to achieve goal (Un)certainity: delegator has access to “one-sample” from the solution space		- Egalitarian → giving and taking agency - Egalitarian, continued: Shared accounts - Joint Attention - Common ground updated through com Goal or information updated (goal can l but information asymmetry needs to be - goal of the collaborators should be to n expected utility of joint decision

Goal			
Is collaborative communication required?	No	Yes	Yes
Actors and their power	- There is a verifier and a solver - in particular, delegator = verifier, delegatee = solver - The delegatee optimizes for the guidelines provided or rubric provided by the verifier		- There is a shared verifier and solver - Both parties assume the roles of verifier and solver - The policy is to collectively arrive at a decision that both parties think is good; the objective is to reach an objectively good decision
Participants' Behaviors observed	Delegatee: - Asks questions to maximize understanding - Proposes solutions Delegator: - Evaluates solutions - Gives good feedback (only based on the pre-existing rubric)		H1 collaborates with H2 Both exchange information (What is that in and how is that conveyed) ---- Both are individuals: What roles do they assume? do the roles change over time? For the collaboration to be successful what kind of properties should they have? ---- ----- Information flow (i.e. what info to convey): - The goal potentially gets updated - There is an information exchange - No party dominates the entire conversation Follows principles of cooperative communication Grice's maxims (how do you convey info): - Clarifications - Acknowledgements - followup: elaboration on a prior utterance (implicitly signal understanding of u). - Repair Agency - Both parties should have their own optimization objectives [Ego] ; Both parties should be able to probe the other's optimization objectives during conversation - Both parties should have agency; Taking away giving power should both be possible. Each party should have a sense of self; Can they convey to the other Actions agents take: - Acting on memory - Solution proposals - Giving good feedback - Come to a consensus Good feedback - Repair Violations of grice's maxims only when they reach a better objective or for repair

Communication principle 2			
Communication principle 3			

Is collaborative AI the same as collaboration?

- Fei
- ☒ read "Completion != Collaboration"
- ☐ what's missing from their framing / what's the blind spots in their framing that we are addressing
- ☐ be (more) aware of our blind spots
- Nikhil
- ☐ Write first draft on the position

Literature Review

[@Fei Fang]
the Rifts paper came in handy. Below is screenshot from section 7: Discussion and Conclusion

Towards LLMs that initiate grounding. Decision-theoretic methods could guide when and how LLMs initiate grounding actions, based on inferred uncertainties in mutual understanding (see Horvitz (1999); Mozannar et al. (2024)). Instruction tuning could be revised to incorporate grounding, and our forecaster could serve as a reward model in RLHF (Ouyang et al., 2022). System prompts and dialogue management show promise, including prompts to disambiguate user intentions (Chen et al., 2023).

Horvitz (1999) is "Principles of mixed-initiative user interfaces" (CHI'99) (and was cited by "Completion != Collaboration" as well). Encouragingly, in this paper, decision-theoretic framing meets mixed-initiative; it also draws a distinction between agent vs. user.

FROM BELIEFS TO ACTIONS
Given uncertainties about a user's goals, what automated actions should be taken? We shall consider the case of a decision about whether or not to invoke the services performed by an intelligent agent. From the perspective of decision theory, decisions about action versus inaction should be directed by expected utility. Autonomous actions should be taken only when an agent believes that they will have greater expected value than inaction for the user,

	Desired	Goal	Not	Desired
Action	$u(A, G)$		$u(A, \neg G)$	
No Action	$u(\neg A, G)$		$u(\neg A, \neg G)$	

Table 1. Four outcomes considered in decisions about whether to engage an intelligent agent to provide service.

Mozannar et al: When to Show a Suggestion? Integrating Human Feedback in AI-Assisted Programming

- Horvitz is last author on this. From a quick glance, this is instantiating the 1999 framework in a particular HAI application, so it might be good reference for us as we think about operationalization.

Chen at al. is written by (your friend) Max Chen; worth looking at their eval for due diligence: Controllable Mixed-Initiative Dialogue Generation through Prompting

[@Nikhil Sharma]

[Angle of inter-personal communication](#)

Put more abstractly, interpersonal communication study seeks to answer these kinds of questions: Does someone who ignores you communicate anything to you? How can the same word or phrase have such different meanings to people? How can people talk anyway? Are arguments and disagreements really summarized in the now-famous line from the film 'Cool Hand Luke': 'What we have here is a failure to communicate'? The difficulty for interpersonal communication has been to move from the anecdotal to the general.

Communication is the process to lower the entropy of the communication and the goal of the communication only gets clear over time and does not necessarily need to be clear from the beginning

'In order for a fidelity assessment to work, one has to assume a considerable degree of both discreteness and stability in the phenomena being matched or fitted' (p. 370). The sender and receiver proposed by the model must then have access to their own and each other's goals and messages so that they can compare them and assess fidelity. Postmodernist thought asks whether humans do indeed assess the fidelity of claims in this way or whether communication is rather an ongoing process that comes clear only over time.

communicate. People understand through communication, not prior to it.

[To communicate one must have a sense of self and thus have power to steer the conversation or give that power to the user]

power. Mainline interpersonal work treats power as 'a communicative product or process through which one person effects an intended behavioral or attitudinal change in another person' (p. 184); the alternative view treats power from the systemic level. These three factors blind us to the ideological basis of interpersonal domination.

Lannamann identifies four ideological characteristics which suggest trends in interpersonal communication research. First, the research tradition tends to select the individual as the locus of personhood. Empirical instruments (questionnaires, content analysis, interviewing) focus on the individual, even when theorists acknowledge that interpersonal communication takes place *between* individuals. Such an approach ignores the social origins of the self (p. 187) and shifts explanations to

[\[Logic and Conversation\]](#)

communication is not merely a transfer of data packets but is essentially a cooperative act where participants tacitly agree to contribute to the exchange in a way that facilitates the mutual goal

[\[Language Models in Dialogue Conversations - adding to Grice's Maxims\]](#)

Benevolence: This maxim requires that the agent not only be truthful and relevant but also actively avoid harm and refuse to engage in unethical requests. In human collaboration, this is often implicit in the social contract; for AI, it must be an explicit constraint.

Transparency: This maxim addresses the "Black Box" problem of AI delegation. It requires the agent to recognize and state its own knowledge boundaries (e.g., "I don't know") and operational capabilities. This directly addresses the need for "metaknowledge" in delegation—a theme we will return to with Minsky and Baird & Maruping.

[\[Origins of Human Communication \(2008\)\]](#)

- **Joint Attention:** This is the cognitive precursor to collaboration. It is the ability to focus on an object *with* another person, knowing that they are also focusing on it. A child pointing at a bird is not just indicating the bird; they are aligning their mental state with the parent. This triangulation (Self-Other-Object) creates a "shared space" of perception.

- **Common Ground:** This is the shared mental workspace of knowledge, beliefs, and values. Collaboration is impossible without a recursive understanding of common ground ("I know that you know that I know..."). It is the database of shared assumptions that allows Gricean implicature to function.

Tomasello draws a sharp distinction between true collaboration and "parallel activity" or "social tool use." Chimpanzees may hunt in groups... chasing the same colobus monkey, but they do not form a "joint goal" in the human sense. Each chimp is maximizing its own chances; i.e. if one chimp disappears or the monkey escapes, the others simply move on. There is no sense of "we" are hunting.

In contrast, human collaboration involves a "joint commitment." If one partner fails, the other attempts to re-engage them or help them, signaling a commitment to the joint activity rather than just the individual reward. This distinction is vital for our taxonomy: **Collaboration** requires a commitment to a joint goal and the formation of a "We" agent. **Delegation**, often, is closer to "social tool use"

[Society of mind by Minsky is delegation hierarchy vs societies of mind where cross agency communication is required for more open ended tasks]

Human-AI dynamics: are we **"giving"** agency to AI (**delegation**), or is the AI **"taking"** agency (**autonomous behavior**), and how does that shift the power balance? - wild chat (invivo chat) - break down into tasks and run analysis and see if power dynamics change (baseline-human)

[TOM]

understanding the other not as an object of observation (3rd person) but as an interacting partner (2nd person)

[Model-Generated](#)

Collaboration is a cooperative process where two or more individuals or groups work together as equals, **sharing responsibility**, authority, and decision-making power to achieve common goals. In collaborative arrangements, participants contribute their unique skills and perspectives while maintaining equal standing in the working relationship. Delegation, in contrast, is a hierarchical process where a person in authority transfers specific tasks, responsibilities, or decision-making power to a subordinate while **retaining ultimate accountability** for the outcomes. The delegating party maintains oversight and control over the process, establishing clear expectations and boundaries for the delegated work

[Barattucci et al., 2020](#)

Delegation operates within clear hierarchical frameworks where authority is transferred from superior to subordinate, with the delegating party maintaining ultimate oversight

[Vall et al., 2019](#)

In contrast, collaboration functions as an egalitarian process where participants work as equals without traditional leader-subordinate relationships

<https://www.semanticscholar.org/p/271398310>

Effective delegation requires clear communication of responsibilities, roles, and expectations to prevent employee frustration and ensure alignment with organizational objectives

[Roe et al., 2025](#)

For delegation to succeed, leaders must possess the individual capability to recognize when delegation is not feasible and maintain close relationships that enable them to rectify discrepancies as they arise

Delegation vs evaluation - do I have a vested interest in the answer being correct

Paper Overview

- ☐ Introduction
- ☐ Related Works
- ☐ Evaluation
- ☐ Method
- ☐ Results
- ☐ Discussion